

## ARIPEKA, FL, USA

WMO: 994082

Lat: 28.4331N

Lon: 82.6667W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |            |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF        |
|---------------|-------------|-------------|-------------------------------|------------|------------|------------|------------|------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|------------|
|               |             |             | 99.6%                         |            |            | 99%        |            |            | 0.4%                  |             | 1%          |             |                       |           |            |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP         | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |            |
| (a)<br>1      | (b)<br>38.8 | (c)<br>42.3 | (d)<br>N/A                    | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>23.8           | (k)<br>59.0 | (l)<br>19.8 | (m)<br>61.0 | (n)<br>4.4            | (o)<br>70 | (p)<br>N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |            |            |            |            |            | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|------------|------------|------------|------------|------------|----------------------|------------|
|               |                        | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |            | 1%         |            | 2%         |            |                      |            |
|               |                        | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB       | WB         | MCDB       | WB         | MCDB       | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>11.8            | (c)<br>95.4     | (d)<br>N/A | (e)<br>93.5 | (f)<br>N/A | (g)<br>91.0 | (h)<br>N/A | (i)<br>N/A          | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>7.4           | (p)<br>300 |

| Dehumidification DP/MCDB and HR |            |            |            |            |            |            |            |            | Enthalpy/MCDB |            |            |            |            |            | Extreme<br>Max WB |
|---------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|---------------|------------|------------|------------|------------|------------|-------------------|
| 0.4%                            |            |            | 1%         |            |            | 2%         |            |            | 0.4%          |            | 1%         |            | 2%         |            |                   |
| DP                              | HR         | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth          | MCDB       | Enth       | MCDB       | Enth       | MCDB       |                   |
| (a)<br>N/A                      | (b)<br>N/A | (c)<br>N/A | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A    | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A | (p)<br>N/A        |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |           |            |           |  |  |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|-----------|------------|-----------|--|--|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |           | n=50 years |           |  |  |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max       | Min        | Max       |  |  |
| (a) 18.5          | (b) 15.1 | (c) 12.8 | (e) 34.4                   | (f) 94.4 | (g) 3.6            | (h) 3.6 | (i) 31.8                                           | (j) 97.0 | (k) 29.8   | (l) 99.1 | (m) 27.8   | (n) 101.1 | (o) 25.2   | (p) 103.7 |  |  |
|                   |          |          | DB                         |          |                    |         |                                                    |          |            |          |            |           |            |           |  |  |
|                   |          |          | WB                         |          |                    |         |                                                    |          |            |          |            |           |            |           |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 72.2   | 59.8  | 61.8  | 65.1  | 71.2  | 76.5  | 80.7  | 82.2  | 82.6  | 80.8  | 74.9  | 67.0  | 63.3  |
|                                                                                  | DBStd       | 10.20  | 8.83  | 8.84  | 7.30  | 5.09  | 4.37  | 3.41  | 3.07  | 2.73  | 3.38  | 6.19  | 6.67  | 8.84  |
|                                                                                  | HDD50       | 43     | 21    | 9     | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 10    |
|                                                                                  | HDD65       | 667    | 208   | 150   | 92    | 11    | 0     | 0     | 0     | 0     | 0     | 9     | 56    | 141   |
|                                                                                  | CDD50       | 8144   | 324   | 340   | 472   | 636   | 820   | 920   | 998   | 1009  | 923   | 771   | 509   | 422   |
|                                                                                  | CDD65       | 3294   | 47    | 61    | 96    | 197   | 355   | 470   | 533   | 544   | 473   | 315   | 115   | 88    |
|                                                                                  | CDH74       | 31200  | 222   | 381   | 430   | 1259  | 3329  | 5164  | 6007  | 6338  | 4716  | 2486  | 523   | 345   |
|                                                                                  | CDH80       | 11373  | 37    | 119   | 57    | 288   | 994   | 2011  | 2538  | 2695  | 1811  | 689   | 86    | 48    |
| Wind                                                                             | WSAvg       | 5.7    | 5.5   | 5.3   | 6.1   | 6.3   | 6.4   | 5.9   | 5.3   | 5.9   | 5.2   | 5.4   | 5.4   | 5.2   |
| Precipitation                                                                    | PrecAvg     | 51.3   | 2.7   | 3.0   | 3.0   | 2.4   | 1.8   | 7.3   | 8.5   | 8.0   | 6.7   | 3.3   | 1.5   | 2.7   |
|                                                                                  | PrecMax     | 63.9   | 5.5   | 9.6   | 7.4   | 5.9   | 8.4   | 15.2  | 12.3  | 12.1  | 13.5  | 8.3   | 4.7   | 12.2  |
|                                                                                  | PrecMin     | 40.2   | 0.8   | 0.3   | 0.3   | 0.2   | 0.2   | 1.4   | 5.0   | 4.3   | 1.7   | 0.1   | 0.4   | 0.6   |
|                                                                                  | PrecStd     | 6.9    | 1.4   | 2.4   | 2.0   | 1.5   | 1.8   | 3.0   | 2.4   | 2.0   | 3.3   | 1.9   | 1.2   | 3.5   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 84.4  | 88.7  | 84.9  | 91.3  | 94.9  | 96.8  | 96.9  | 96.6  | 96.6  | 94.6  | 86.8  | 85.6  |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 2%          | DB     | 79.9  | 84.0  | 81.9  | 86.4  | 91.8  | 93.2  | 95.1  | 93.9  | 94.2  | 89.7  | 82.3  | 80.7  |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 5%          | DB     | 75.9  | 78.3  | 79.1  | 83.2  | 88.6  | 90.2  | 91.9  | 90.8  | 90.4  | 85.9  | 79.4  | 77.9  |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 10%         | DB     | 72.2  | 73.8  | 76.4  | 80.2  | 85.0  | 87.4  | 88.4  | 88.6  | 87.4  | 83.7  | 76.9  | 75.1  |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.9  | 15.5  | 15.4  | 15.0  | 13.7  | 11.8  | 12.1  | 11.8  | 12.8  | 14.4  | 15.7  | 15.5  |
|                                                                                  |             | MCDBR  | 18.3  | 19.2  | 18.2  | 18.9  | 17.8  | 14.7  | 15.9  | 14.3  | 16.0  | 15.3  | 17.0  | 16.2  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.344 | 0.356 | 0.367 | 0.391 | 0.426 | 0.455 | 0.482 | 0.464 | 0.423 | 0.389 | 0.353 | 0.341 |
|                                                                                  | taud        |        | 2.516 | 2.490 | 2.448 | 2.351 | 2.310 | 2.306 | 2.236 | 2.301 | 2.385 | 2.459 | 2.505 | 2.549 |
|                                                                                  | Ebn at Noon |        | 285   | 291   | 293   | 287   | 276   | 266   | 259   | 263   | 272   | 275   | 279   | 281   |
|                                                                                  | Edh at Noon |        | 29    | 32    | 35    | 40    | 42    | 42    | 45    | 42    | 37    | 32    | 29    | 27    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 1098  | 1318  | 1688  | 2035  | 2165  | 1949  | 1831  | 1769  | 1613  | 1483  | 1204  | 1012  |
|                                                                                  | RadStd      |        | 77    | 92    | 150   | 114   | 124   | 153   | 111   | 111   | 113   | 96    | 69    | 85    |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (33 neighbors) |  | +0.77 | N/A     | N/A    | N/A     | N/A   | N/A   | -9          | -92   | +295  | +207  |

Nomenclature: See separate page